2023-27

Probability → CSE → 2nd sem

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SETS

A Set is a well-defined collection of objects, which are not repeated.

Eg;

let

(i) A = { 1, 2, 3, 4, 5 } → Yes

(ii) B = { 1, 2, 3, 4, 3, 5 } → No

Because ⇒ 3 is repeated one times.

(iii) Collection of vowels → Yes { a, e, i, o, u }

(iii) Collection of all intelligent student of VVIT CSE 2023-27 ⇒ No

Because ⇒ no well-defined collection

Symbol :-

(i) { } → braces → all element ~~rep~~ listed between this brace.

(ii) 'ϵ' → belongs to

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*→ Important in probability

→ Types of set :-

(i) Empty set :- (or null set)

A set having no element.

It is denoted by ϕ

Eg;

let A = set of even number not divisible by 2.

Here A is empty set.

> Note:- The set {0} is not an empty set as it contain one ~~tem~~ element 0.

(ii) Equivalent set* :-

Two finite set A and B are ~~set~~ said to be equivalent if they have same number of element in both set.

It is denoted by n(A).

let A = {1, 2, 3, 4, 5} and B = {a, e, i, o, u}

n(A) = n(B)

∴ set A and set B are equivalent.

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Note:-
$n(A) \rightarrow$ Number of elements in A.

(iii) Universal Set :-

Any set which is super set of all sets under consideration is called the universal set.

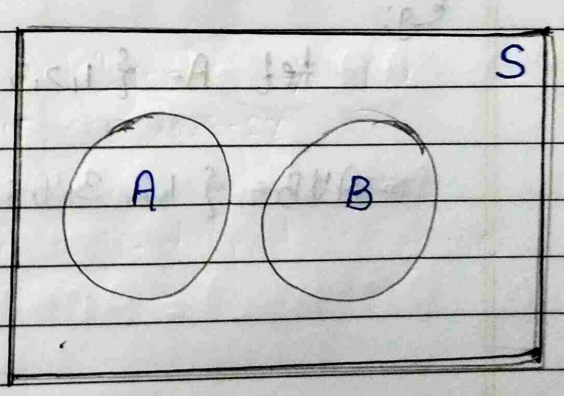
It is denoted by U or S or Ω.

Eg; let $A = \{1,2,3\}$, $B = \{3,4,5,6,7\}$, $C = \{5,6,7,8,9\}$

$S = \{1,2,3,4,5,6,7,8,9\}$

→ Venn diagram* :-

Here,
S = universal set and,
A and B are set.



(i) Union of sets :-

The union of two sets A and B is the set of all those element which are either in A or in B or in Both.

It is denoted by 'U'.

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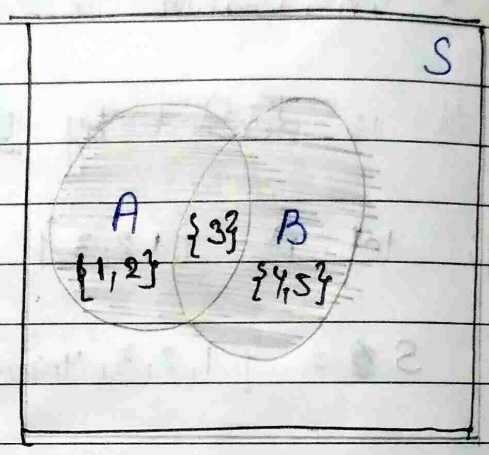
$A \cup B = \{ A \ \underline{or} \ B \}$

Eg:

let $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$

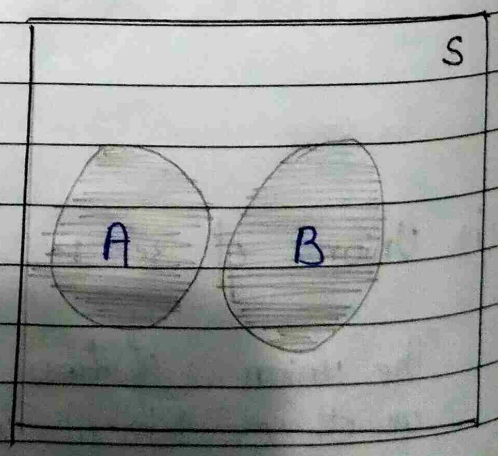
$A \cup B = \{1, 2, 3, 4, 5\}$



Eg:

let $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$

$A \cup B = \{1, 2, 3, 4, 5, 6\}$



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(ii) Intersection of sets :-

The intersection of two sets A and B ~~are~~ is the set of all those element which are common in A and B.

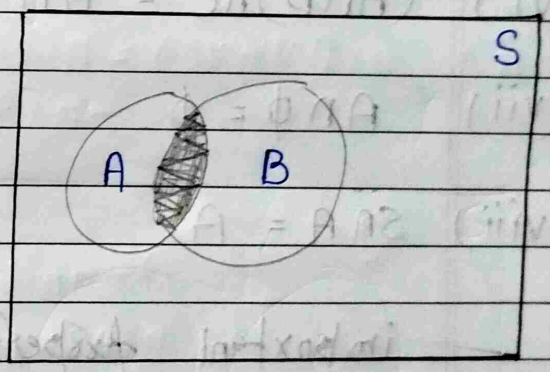
It is denoted by '$\cap$'.

$A \cap B = \{A \text{ and } B\}$

Eg; let

$A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$

$A \cap B = \{3\}$

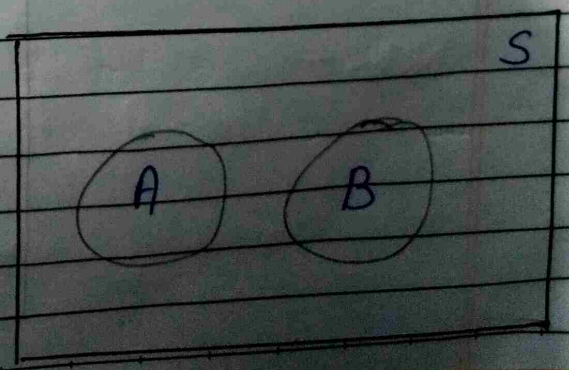


Eg;

let

$A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$

$A \cap B = \phi$



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→ Properties of Union and intersection* :-

(i) $A \cup B = B \cup A$

(ii) $(A \cup B) \cup C = A \cup (B \cup C)$

(iii) $A \cup \phi = A$

(iv) $S \cup A = S$ [Universal set → S]

(v) $A \cap B = B \cap A$

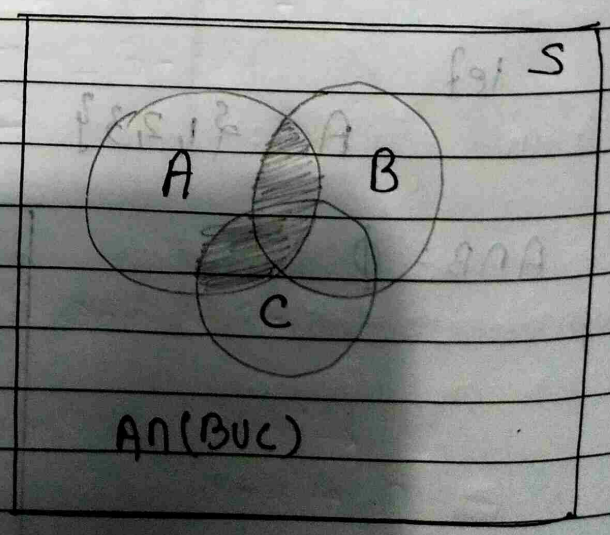
(vi) $(A \cap B) \cap C = A \cap (B \cap C)$

(vii) $A \cap \phi = \phi$

(viii) $S \cap A = A$

important properties :-

(i) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$



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(iii) Complement of a set :-

The complement of a set A is the set of all those element of the universal set S which are not element of A.

It is denoted by A'.

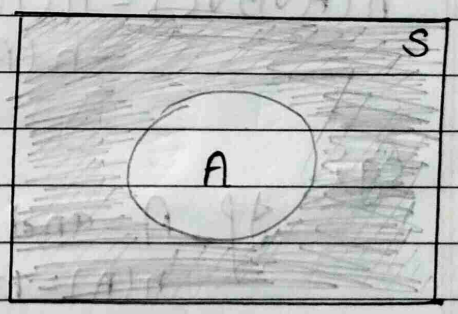
Eg;

let

$S = \{1, 2, 3, 4, 5\}$

$A = \{3, 4, 1\}$

$A' = \{2, 5\}$



→ Properties :-

(i) $A \cap A' = \phi$

(ii) $A \cup A' = S$

(iii) $\phi' = S$

(iv) $S' = \phi$

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(v)* $(A \cup B)' = A' \cap B'$

(vi)* $(A \cap B)' = A' \cup B'$

(vii) $(A')' = A$

→ * Some theorem of number of elements of set :-

(i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

(ii) $n(A \cap B) = n(A) + n(B) - n(A \cup B)$

(iii) $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$.

Eg;

If A and B are two set such that $n(A) = 17$, $n(B) = 23$ and $n(A \cup B) = 38$ find $n(A \cap B)$?

$n(A \cap B) = n(A) + n(B) - n(A \cup B)$
$= 17 + 23 - 38$
$n(A \cap B) = 2$

Note:-

Use of $\cup$ and $\cap$:-

(i) check if written 'or' → $\cup$
'and' → $\cap$

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(ii) check if written 'both' → $\cap$

check if written 'either of these two or neither both' → $\cup$

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